

SENTINEL-CPS

Thesis Synopsis

Securing Cyber Physical Systems in a Remote IoT Lab: A Zero-Trust Edge Gateway Architecture Utilizing eBPF and AI Threat Detection

Project Name: Sentinel-CPS

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Primary Focus	Secure remote CPS education, Zero-Trust gateway mediation, and eBPF/AI threat monitoring

Prepared for Master's Thesis Development

Document Control

This synopsis updates the approved thesis concept to reflect the current Sentinel-CPS implementation direction: a remote educational sandbox, a hardened Zero-Trust Gateway, a TV-as-a-Track cyber-physical execution layer, and planned host-level eBPF/AI threat monitoring.

Version	Date	Prepared By	Summary of Update
10.0	May 29, 2026	Tarique Chowdhury	Reframed the thesis synopsis around the Sentinel-CPS v10.0 architecture while preserving the approved thesis title; clarified the Bastion Host educational sandbox, Smart TV track substrate, light-sensor vehicle navigation, encrypted ESP-NOW telemetry, and eBPF/AI mitigation pipeline.

1. Abstract / Synopsis

The rapid integration of Cyber-Physical Systems (CPS) into university curricula creates a critical security challenge: how to provide remote, potentially untrusted undergraduate students with hands-on access to IoT hardware without exposing the broader enterprise network to lateral movement, unauthorized control, or localized physical hijacking. This thesis presents Sentinel-CPS, a dual-layer architecture that merges a secure remote educational sandbox with a hardened Zero-Trust execution environment for autonomous IoT experimentation.

Sentinel-CPS supports secure student learning through a restricted Windows Bastion Host reached through the university VPN. This access model allows remote users to interact with isolated development hardware while preventing direct exposure of the core CPS control plane. In parallel, the Sentinel Gateway - a bare-metal Raspberry Pi - enforces strict Layer 2 Hardware-Level Access Control (HLAC) through nftables and key-based administrative access. The Gateway supervises a TV-as-a-Track physical environment in which autonomous ESP32 edge vehicles use downward-facing analog light sensors to navigate a dynamic digital track rendered on a Smart TV display.

The central research contribution is the integration of host-level eBPF kernel tracing with lightweight AI inference to monitor the command and telemetry path associated with encrypted ESP-NOW edge communication and the Gateway serial chokepoint. By combining application logging, kernel-level read/write tracing, baseline behavioral modeling, and emergency STOP/severance logic, Sentinel-CPS demonstrates a verifiable method for detecting and mitigating data injection, replay, and denial-of-service conditions at the edge while preserving the physical safety and network integrity of remote IoT educational laboratories.

2. Research Problem

Remote IoT and robotics laboratories are valuable for cybersecurity and computer science education, but they introduce risk when physical devices are made accessible from outside the lab. Traditional remote lab designs often expose control interfaces, development hosts, or edge devices in ways that create opportunities for unauthorized access, lateral movement, unsafe actuator behavior, and weak visibility into low-level runtime activity.

- **Network risk:** Remote access can unintentionally create pathways from student-facing systems into the broader university network.
- **Physical risk:** Autonomous devices can be hijacked, flooded, or driven into unsafe states if control traffic is spoofed or abused.
- **Observability gap:** Application logs alone may not reveal low-level command injection, abnormal serial activity, or telemetry flooding at the host boundary.
- **Educational constraint:** The system must still remain usable for students, instructors, and reviewers without requiring fragile manual lab reconfiguration.

3. Proposed Sentinel-CPS Architecture

The proposed solution separates educational access, gateway enforcement, edge communication, and physical execution into distinct layers. Each layer has a narrow purpose and a defined trust boundary.

Layer	Purpose	Primary Components	Security Role
Remote Educational Sandbox	Allows remote users to safely interact with development hardware.	University VPN, Windows Bastion Host, RDP workflow, isolated microcontrollers.	Concentrates student ingress at a controlled host rather than exposing the Gateway directly.
Zero-Trust Gateway	Mediates commands, logs activity, hosts the web interface, and enforces network policy.	Raspberry Pi 4, systemd, Flask C2 app, nftables HLAC, Ed25519 SSH, eBPF tooling.	Serves as the hardened control and observability chokepoint for the CPS.
Encrypted Edge Telemetry	Transports operational telemetry and motor commands between the Gateway Hub and vehicles.	Tethered ESP32-D0WDQ6 Hub, ESP-NOW, CCMP encryption, ESP32 vehicle nodes.	Reduces direct wireless exposure and constrains edge communication through an authorized bridge.
TV-as-a-Track Execution Layer	Converts the digital command interface into a physical navigation substrate.	Smart TV Display Node, Lane Builder, digital twin overlay, analog light sensors, PID control.	Enables repeatable physical experiments without exposing the edge nodes to the enterprise network.

4. Primary Research Contributions

- **Zero-Trust remote CPS education model:** The thesis defines an access pattern that allows remote educational use while keeping the Gateway and edge devices shielded from direct uncontrolled exposure.
- **TV-as-a-Track cyber-physical interface:** The Smart TV becomes both the dynamic physical road surface and the digital twin display, enabling rapid track changes without manual reconstruction.
- **Light-sensor autonomous navigation:** Downward-facing analog sensors replace camera-dependent navigation to reduce latency, bandwidth demand, and visual spoofing surface.
- **Kernel-level serial observability:** eBPF tracing targets the host-level read/write behavior associated with `/dev/ttyUSB0`, producing telemetry that is harder to spoof than application-only logs.
- **AI-assisted anomaly detection and mitigation:** A lightweight inference layer evaluates telemetry against expected behavior and supports emergency STOP/severance action when abnormal activity is detected.

5. Expected Validation Focus

1. **Access control validation:** Confirm that only approved endpoints can access SSH or the C2 web application and that the Display Node is not granted administrative privileges.
2. **Remote operation validation:** Demonstrate that users can generate tracks, observe telemetry, and interact with the CPS through the Bastion Host without direct Gateway exposure.
3. **Physical navigation validation:** Measure whether ESP32 edge vehicles can reliably follow TV-rendered tracks using analog light sensors and onboard PID control.
4. **Observability validation:** Collect and compare application logs, serial I/O traces, and edge telemetry to establish nominal system behavior.
5. **Threat mitigation validation:** Simulate injection, replay, or DoS-like traffic conditions and evaluate whether the Gateway can detect abnormal behavior and issue a safe STOP/severance response.

6. Expected Outcome

The expected outcome is a functional, thesis-ready prototype demonstrating that a remote CPS education lab can support realistic IoT and autonomous robotics experimentation while maintaining strict Zero-Trust boundaries. Sentinel-CPS aims to show that remote access, physical autonomy, dynamic test environments, and host-level security monitoring can coexist in a compact educational architecture.

built from accessible hardware such as Raspberry Pi, ESP32 microcontrollers, and a Smart TV display substrate.

If successful, the work will provide a reusable architectural pattern for universities that want to teach hands-on CPS, IoT, and cybersecurity concepts without exposing unmanaged physical hardware directly to institutional networks or remote users.